

HARIS WAHEED BHATTI

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Resident in Slovenia with full-time work rights; post-study residence permit from 09/2026. EU Blue Card eligible.

PROFILE

Software-focused engineer building tooling for the AEC industry, working at the intersection of IFC schemas, graph databases, and LLM agent systems. Author of a published open-source Python package for IFC-to-Neo4j conversion; MSc research benchmarked graph-mediated LLM retrieval against a published direct-IFC baseline.

SKILLS

Programming & Data: Python, TypeScript, Solidity, Neo4j, Cypher, knowledge graphs, Git, LaTeX
AI / LLM: Agentic systems, Model Context Protocol (MCP), RAG, LLM evaluation and benchmarking
BIM & AEC: IFC (IFC2X3, IFC4, IFC4X3), IfcOpenShell, Linked Building Data ontologies, Revit, Dynamo, Blender, Autodesk Construction Cloud, Three.js, web-ifc
Languages: Urdu (native), English (fluent), Slovene (A1)

PROJECTS

- ifctoneo4j** – Python package (PyPI) | *Python, IfcOpenShell, Neo4j, Cypher* 2026
- Open-source library converting IFC models (IFC2X3, IFC4, IFC4X3) into **Neo4j** property graphs aligned to Linked Building Data ontologies, with no RDF intermediate. Published and installable via `pip install ifctoneo4j`
- BIMGraphExplorer** – LLM agent for IFC question answering | *Python, MCP, Neo4j* 2026
- Adaptive querying system exposing four read-only tools over the Model Context Protocol; the model profiles the graph schema and issues iterative Cypher queries conditioned on prior results
 - Ran a controlled benchmark against a published direct-IFC baseline at matched model, matched validated judge, and matched scope (23 IFC datasets, 367 questions, 15 projects): 72.0 % versus 82.3 % accuracy, transparency at parity (99.7 % versus 100 %), and outright wins on 5 of 23 datasets
 - Raised accuracy by 33 percentage points on a fixed model and dataset through prompt design alone, lifting source-transparency from 12.5 % to 99.7 %
- Blockchain-Based BIM Document Management** | *Solidity, Ethereum (Arbitrum Sepolia)* 2024
- Smart contract system to digitally sign and manage construction documents, with off-chain storage linked on-chain for immutability, designed to streamline dispute resolution under **FIDIC** contracts; funded by a competitive PKR 100,000 grant from the Pakistan Engineering Council
- Federated BIM Model** – Pakistani-Themed Expo, Guimarães | *IFC, Autodesk Construction Cloud* 2024
- Led BIM coordination across architecture and MEP disciplines; delivered a federated model with clash detection and information-requirement compliance

EDUCATION

- European Master in Building Information Modelling (BIM A+)** 10/2025 – 07/2026
University of Ljubljana, UL FGG Slovenia
- Field of study: Information Management in Construction; completed in 10 months via an accelerated thesis track
 - Thesis:** *Graph-Mediated LLM Exploration for IFC Question Answering: A Benchmark Comparison Against Direct-IFC Approaches*. Supervisor: Prof. Žiga Turk
- BE Civil Engineering** 09/2020 – 06/2024
National University of Sciences and Technology (NUST) Pakistan
- CGPA: 3.25/4.00

EXPERIENCE

- Part-time Employment** 02/2025 – 07/2026
Ljubljana, Slovenia
- Worked evenings and weekends throughout MSc studies to self-fund living costs in Slovenia
- Contract Management Intern** 08/2023 – 09/2023
Engineering and Management Services Pakistan
- Reviewed contract reports and arbitration documentation under **FIDIC** conditions; presented findings to the internal legal team
- Research Intern (Remote)** 07/2023 – 08/2023
Southern Illinois University Edwardsville USA
- Synthesised academic literature to support a faculty-mentored research proposal
- Civil Engineering Intern** 07/2023 – 08/2023
Fatima Fertilizer Company Pakistan
- Modelled a 6,804 m² rock phosphate storage shed in 3D from 2D drawings, coordinating with stakeholders on design refinement